

SLIM TRABELSI

Mechanical Engineering Student

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PROFILE

Mechanical engineering student at ENIT, with a track record of **6+ technical projects** delivered and **3 professional experiences** in industrial environments. Contributed to improving the optical efficiency of a CPC solar concentrator by **+15%**, developed and programmed **2 autonomous robots** within the ENIT Robotics Club, and gained hands-on experience in manufacturing, welding, and industrial maintenance at **Future Pub** and **GFCO** and turned his focus toward industrial systems optimization at **Marquardt**. Results-driven, autonomous, and capable of leading projects from design through to completion.

PROFESSIONAL EXPERIENCE

- **Marquardt** Tunisia
Engineering Intern – Facility Management Summer 2026
 - **Data Analysis:** Conducted in-depth analysis of operational data and KPIs for the compressed air system.
 - **Field Diagnosis:** Performed on-site diagnosis of the compressed air network to assess performance and identify variability sources.
 - **Continuous Improvement:** Identified root causes of energy performance fluctuations and proposed corrective action plans.
- **Future Pub – Metal Structures, Fabrication & Design** Tunisia
Part-Time Job 2020 – Present
 - **Fabrication:** Fabrication and assembly of metal structures .
 - **Machining:** MAG welding, turning, and milling operations on metal components.
 - **Installation:** On-site assembly and installation, managing field constraints and deadlines.
 - **CAD:** Design and 3D modeling of structures using SolidWorks prior to fabrication.
- **Grande Fabrique De Confiserie Orientale (GFCO)** Tunisia
Industrial Internship Summer 2025
 - **Maintenance:** Exposure to industrial maintenance department operations and workflows.
 - **Operations:** Participated in preventive and corrective maintenance tasks.
 - **Organization:** Observed planning and organizational methods in an industrial environment.

ACADEMIC EXPERIENCE & PROJECTS

- **ENIT Robotics Club** Tunis, Tunisia
Active Member & Bureau Officer 2025 – 2027
 - **Active Member:** Designed an all-terrain robot and a line-following robot; handled wiring, coding, fabrication, and competed in national competitions.
 - **Bureau Officer:** Organized technical training sessions, mentored members, and coordinated a national robotics event.
- **PFA – Modeling & Techno-Economic Evaluation of a Next-Gen Solar Concentrator (CPC)**
Final Year Project | Python, JavaScript, Tonatiuh, SolidWorks
 - **Optimization:** Geometric optimization of the CPC concentrator, improving optical efficiency by **+15%**.
 - **Simulation:** Optical simulation using Tonatiuh across **20+ configurations** to analyze solar flux distribution.
 - **Modeling:** Full CAD modeling of the system, reducing prototyping time by **30%**.
 - **Tools & Web:** Developed performance calculation scripts and a web interface to test different design variables.
- **Structural Analysis – FEM**
Academic Project | Abaqus, SolidWorks
 - **FEA:** Performed finite element analysis on mechanical components using Abaqus.
 - **Results:** Modeled geometry in SolidWorks and analyzed stress, strain, and deformation outputs.
- **Electric Wood-Cutting Saw – BE Project**
Engineering BE Project | SolidWorks, Mechanical Design
 - **Design:** Designed and built an electric saw for wood cutting using a high-torque motor; modeled the full frame in SolidWorks.

EDUCATION

- **National Engineering School of Tunis (ENIT)** Tunis, Tunisia
Engineering Degree – Mechanical Engineering *2024 – Present*
 - **Key Modules:** Thermodynamics, Fluid Mechanics, Mechanical Design, Finite Element Analysis (FEA), Manufacturing, Industrial Maintenance.
- **El Manar Preparatory Institute for Engineering Studies (IPEIEM)** Tunis, Tunisia
Preparatory Classes – Physics & Technology (PT) *2022 – 2024*
- **Technical Baccalaureate** Tunisia
With Honors *2022*

TECHNICAL SKILLS

- **CAD / Simulation:** SolidWorks, CATIA, Abaqus, Ansys, Tonatiuh, MATLAB.
- **Programming:** Python, C/C++, Arduino, JavaScript, HTML/CSS, Pascal.
- **Engineering Knowledge:** Thermodynamics, Fluid Mechanics, Mechanical Design, FEA, System Maintenance & Reliability.
- **Soft Skills:** Analytical thinking, problem-solving, teamwork, leadership, project management, technical communication.

LANGUAGES

- **English:** TOEIC B2 · **French:** DELF B2 · **Arabic:** Native